Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc2`; Mathlib at commit `1b1ccf4b`; physlib (`master`) and TauCeti (commit `e7e7f52`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 The Born rule is equivalent to the von Neumann entropy

Files.

- File: `briefs/BornRuleEntropy/BornRuleEntropy.lean`
- Library: `BornRuleEntropy`
- Imports: `ForMathlib.Analysis.Convex.AffineOnBall`, `ForPhyslib.QuantumMechanics.Kadison`, `ForPhyslib.QuantumMechanics.Qubit.BlochBall`, `ForPhyslib.QuantumMechanics.VonNeumannEntropyMixture`
- New `ForMathlib` material: `ForMathlib/Analysis/Convex/AffineOnBall.lean`: a real function on a real normed space that is affine along every chord of the unit sphere is affine on the closed unit ball (`exists_linearMap_eq_of_chord`); in an inner product space, if moreover $f \le 1$ on the sphere, $f(a) = 1$ and $f(-a) = 0$ for a unit vector a, then $f(v) = (1 + \langle a, v\rangle)/2$ (`eq_of_chord_of_le_one`).
- New `ForPhyslib` material:

– `ForPhyslib/QuantumMechanics/VonNeumannEntropyMixture.lean`: a mixture $\sum_i p_i \left|\psi_i\right\rangle \left\langle\psi_i\right|$ of orthonormal pure states, in any dimension, is a density operator (`isDensityOperator_tsum_smul_proj`) with entropy $\sum_i \eta(p_i)$, $\eta(x) = -x \log x$ (`vonNeumannEntropy_tsum_smul_proj`); the entropy of an equal mixture of two pure states is $h((1 + |\left\langle\psi|\phi\right\rangle|)/2)$, h the binary entropy (`vonNeumannEntropy_mix_proj`); every density operator is such a mixture, of countably many orthonormal pure states, its eigenvectors with nonzero eigenvalue (`IsDensityOperator.exists_tsum_smul_proj`).

– `ForPhyslib/QuantumMechanics/Qubit/BlochBall.lean`: the Bloch ball of the plane spanned by orthonormal ψ, χ in E. The point $r \in \mathbb{R}^3$ is the operator $\frac{1}{2}(\left|\psi\right\rangle \left\langle\psi\right| + \left|\chi\right\rangle \left\langle\chi\right| + r \cdot \sigma)$ (`blochOperator`), affine in r; unit Bloch vectors are the pure states of the plane; the Born probability of two of them is $(1 + a \cdot b)/2$, so orthogonal means antipodal; and every point of the ball is the mixture of the ends of a diameter through it.

– `ForPhyslib/QuantumMechanics/Qubit/Bloch.lean`: every unit vector of $\mathbb{R}^3$ is the Bloch vector of a unit vector of $\mathbb{C}^2$ (`exists_blochVector_eq`).

Setting. E is a complex Hilbert space of any dimension. A pure state is $\left|\psi\right\rangle \left\langle\psi\right|$ (`proj` ψ) for a unit vector ψ, and an ensemble is a density operator (`IsDensityOperator`: positive, trace class, trace one). The entropy S is a function from operators to $[0, \infty]$, and the von Neumann entropy `vonNeumannEntropy` is $\operatorname{tr} \eta(\rho) \in [0, \infty]$ with the natural logarithm, so the brief's "$S = 1$" (one bit) reads $S = \log 2$. The conditional probability is a function p of two operators; $p(\left|\psi\right\rangle \left\langle\psi\right|, \rho)$ is the probability of finding ψ having prepared ρ. The conditions are:

- `EnsemblesAndStates` p (EQST): $p(\left|\psi\right\rangle \left\langle\psi\right|, \rho) \in [0, 1]$ for density operators ρ; $p(\left|\psi\right\rangle \left\langle\psi\right|, t \left|\phi_1\right\rangle \left\langle\phi_1\right| + (1 - t) \left|\phi_2\right\rangle \left\langle\phi_2\right|) = t\, p(\left|\psi\right\rangle \left\langle\psi\right|, \left|\phi_1\right\rangle \left\langle\phi_1\right|) + (1 - t)\, p(\left|\psi\right\rangle \left\langle\psi\right|, \left|\phi_2\right\rangle \left\langle\phi_2\right|)$ for $t \in [0, 1]$; and $p(\left|\psi\right\rangle \left\langle\psi\right|, \left|\phi\right\rangle \left\langle\phi\right|) = 1$ iff $\left|\psi\right\rangle \left\langle\psi\right| = \left|\phi\right\rangle \left\langle\phi\right|$;

- `BornRule` p (BR-BORN): $p(\left|\psi\right\rangle \left\langle\psi\right|, \left|\phi\right\rangle \left\langle\phi\right|) = |\left\langle\psi|\phi\right\rangle|^2$;

- `IsVonNeumannEntropy` S (BR-ENT): $S(\rho) =$ `vonNeumannEntropy` ρ for density operators ρ;

- `MutualExclusivity` S p ([B]MUTEX): for every countable family (ψ_i) of unit vectors, $p(\left|\psi_i\right\rangle \left\langle\psi_i\right|, \left|\psi_j\right\rangle \left\langle\psi_j\right|) = 0$ for all $i \neq j$ iff $S(\sum_i w_i \left|\psi_i\right\rangle \left\langle\psi_i\right|) = \sum_i \eta(w_i)$ for all weights $w_i \geq 0$ with $\sum_i w_i = 1$ (the sum of operators converging in operator norm);

- `CountablyAffine` p (the affinity of the brief's EQST): for every countable family (ψ_i) of unit vectors and weights $w_i \geq 0$ with $\sum_i w_i = 1$, $\sum_i w_i\, p(\left|\phi\right\rangle \left\langle\phi\right|, \left|\psi_i\right\rangle \left\langle\psi_i\right|)$ converges to $p(\left|\phi\right\rangle \left\langle\phi\right|, \sum_i w_i \left|\psi_i\right\rangle \left\langle\psi_i\right|)$.

Results.

Brief	Lean
Proposition 3, eq. (6), $S(\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi) = h((1+\sqrt{\lambda})/2)$, $\lambda = \lvert\langle\psi\vert\phi\rangle\rvert^2$	`vonNeumannEntropy_mix_proj_eq` (from `vonNeumannEntropy_mix_proj`)
Proposition 3, $p(\phi\vert\psi) = \lambda$ iff $S(\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi) = h((1+\sqrt{\lambda})/2)$	`norm_inner_sq_eq_iff`, for $\lambda \in [0,1]$, with p the Born rule
Fig. 1, eq. (4), a point of the Bloch ball is diagonal in the ends of the diameter through it	`Qubit.exists_blochOperator_eq`
Eq. (5), the Born rule is the squared cosine of the half-angle	`Qubit.norm_inner_planeVector_eq_cos`
Eqs. (4)–(5), the equal mixture is $\frac{1+\sqrt{\lambda}}{2}\rho_+ + \frac{1-\sqrt{\lambda}}{2}\rho_-$ with $\rho_+ \perp \rho_-$	`Qubit.exists_mix_proj_planeVector_eq`
Proposition 3, eq. (6), by the geometric diagonalization	`BlochBall.vonNeumannEntropy_mix_proj_eq`
Fig. 2, the entropy is strictly decreasing in the Born probability	`strictAntiOn_entropyOfBorn`
Corollary 7, $\langle\phi\vert\psi\rangle = 0$ iff $S(\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi) = 1$	`inner_eq_zero_iff`, with $\log 2$ for 1
EQST, BR-BORN, BR-ENT, [B]MUTEX	`EnsemblesAndStates`, `BornRule`, `IsVonNeumannEntropy`, `MutualExclusivity`
Theorem 9, BR-BORN and BR-ENT imply [B]MUTEX	`MutualExclusivity.of_bornRule`
Theorem 9, BR-BORN and [B]MUTEX imply BR-ENT	`IsVonNeumannEntropy.of_bornRule`
Theorem 9, BR-ENT and [B]MUTEX imply BR-BORN	`BornRule.of_isVonNeumannEntropy`, over EQST
Theorem 9, BR-ENT and [B]MUTEX imply BR-BORN, on the Bloch ball (Fig. 3)	`BlochBall.BornRule.of_isVonNeumannEntropy`, over EQST
Fig. 3, an affine function on the ball with maximum 1 at a and value 0 at $-a$ is $(1 + a \cdot r)/2$	`eq_of_chord_of_le_one`
Theorem 9	`two_conditions_imply_third`
(used for the last direction) orthogonal states are mutually exclusive	`apply_eq_zero_of_inner_eq_zero`
(used for the last direction) the Born rule on the qubit spanned by ψ and $\chi \perp \psi$	`apply_planeVec_eq`: $p(\psi\vert r\psi + s\bar{\zeta}\chi) = r^2$; from `apply_planeVec` and `apply_halfVec`

Countable affinity and Corollary 8:

Brief	Lean
EQST, affinity for countable mixtures	`CountablyAffine`; it implies the affinity of `EnsemblesAndStates` (`CountablyAffine.affine`, `EnsemblesAndStates.of_countablyAffine`)
Theorem 9 over the brief's EQST	`two_conditions_imply_third_of_countablyAffine`
Corollary 8, every ensemble is a countable mixture $\sum_i w_i \rho_{\psi_i}$ of pure states, and $p(\phi\|\rho) = \sum_i w_i\, p(\phi\|\psi_i)$	`CountablyAffine.exists_hasSum_proj`
Corollary 8, p is fully specified by its values on pure states	`CountablyAffine.eq_of_eq_on_pure`
(not in the brief) with BR-BORN, $p(\phi\|\rho) = \langle\phi\|\rho\|\phi\rangle$	`CountablyAffine.apply_eq_re_inner`

Independence of the conditions (not in the brief), for E with two orthonormal vectors; every model satisfies EQST and `CountablyAffine`:

Statement	Lean
$p(\psi\|\rho) = \langle\psi\|\rho\|\psi\rangle$ satisfies EQST	`bornProb`, `ensemblesAndStates_bornProb`, `bornProb_countablyAffine`
BR-BORN without BR-ENT or [B]MUTEX: $p = \langle\psi\|\rho\|\psi\rangle$, $S = 0$	`exists_bornRule_not_isVonNeumannEntropy`
BR-ENT without BR-BORN or [B]MUTEX: $p = (1 + \langle\psi\|\rho\|\psi\rangle)/2$, S the von Neumann entropy	`exists_isVonNeumannEntropy_not_bornRule`
[B]MUTEX without BR-BORN or BR-ENT: $p = (1 + \langle\psi\|\rho\|\psi\rangle)/2$, $S = 0$ on pure states and ∞ otherwise	`exists_mutualExclusivity_not_bornRule`
the three models together	`conditions_independent`
in dimension at most one, EQST implies BR-BORN	`BornRule.of_rank_le_one`

Differences from the brief.

- E is a Hilbert space of any dimension; separability is not used. [B]MUTEX quantifies over countable index types (in the universe of E). For BR-BORN + [B]MUTEX ⇒ BR-ENT the eigenbasis of ρ is restricted to the eigenvectors with nonzero eigenvalue, a countable family since the eigenvalues are summable.
- Entropies take values in $[0, \infty]$, not in $\mathbb{R}$: the von Neumann entropy and the Shannon entropy of a countable family of weights can be infinite (see "Problems found").
- In `EnsemblesAndStates`, affinity is assumed only for mixtures of two pure states, which is weaker than the brief's countable mixtures, so `two_conditions_imply_third` is stronger than Theorem 9. The brief's countable affinity is `CountablyAffine`; it implies the two-state affinity, and Theorem 9 over the brief's EQST follows (`two_conditions_imply_third_of_countablyAffine`). EQST is used only for BR-ENT + [B]MUTEX ⇒ BR-BORN, and there only through $p(\psi|\psi) = 1$, affinity for two pure states, and $p \geq 0$ on pure states (the algebraic proof) or $p \leq 1$ on pure states (the Bloch-ball proof). The other two directions do not use

EQST. Corollary 8 needs countable affinity: in infinite dimension an ensemble is in general only a countable mixture of pure states.

- The first argument of p is the projection $|\psi\rangle\langle\psi|$, so p depends on the ray of ψ only.
- In BR-ENT + [B]MUTEX ⇒ BR-BORN, [B]MUTEX is used only for pairs of orthogonal states, and only in the direction "Shannon entropy ⇒ mutually exclusive". In BR-BORN + [B]MUTEX ⇒ BR-ENT only the other direction is used.
- Independence of the conditions is not in the brief; its conclusion only says that [B]MUTEX is the one extra requirement. The formalization shows that none of BR-BORN, BR-ENT and [B]MUTEX implies another over EQST, when E has two orthonormal vectors (in dimension at most one, EQST alone implies BR-BORN). The model for [B]MUTEX alone satisfies it in a degenerate way: $p(\psi|\phi) \geq \frac{1}{2}$ for all pure states, so no two states are mutually exclusive, and every mixture of two or more states has entropy 0 or ∞, never a Shannon entropy $\log 2$. In each model the failure of one of the two remaining conditions follows from Theorem 9.
- BR-ENT + [B]MUTEX ⇒ BR-BORN is proved twice. The first proof follows Fig. 3. The states of the plane spanned by ψ and a unit $\chi \perp \psi$ form a Bloch ball, on which $f = p(\psi|\cdot)$ is affine along every chord, since the ends of a chord are pure states; $f \leq 1$, $f = 1$ at ψ and $f = 0$ at the antipode χ. Such a function is $(1 + a \cdot r)/2$ with a the Bloch vector of ψ (`eq_of_chord_of_le_one`): on a unit $e \perp a$ its linear part vanishes, because otherwise f would exceed 1 on the sphere near a. On the sphere $(1 + a \cdot b)/2$ is the Born rule. The second proof is algebraic, in the same qubit. From $f(\psi) = 1$ and $f(\chi) = 0$, the convex identities $|u\rangle\langle u| + s(r-s)|\chi\rangle\langle\chi| = r(r-s)|\psi\rangle\langle\psi| + 2rs\,|e_\zeta\rangle\langle e_\zeta|$ (for $r \geq s$, and symmetrically for $r \leq s$), with $u = r\psi + s\bar{\zeta}\chi$ and $e_\zeta = (\psi + \bar{\zeta}\chi)/\sqrt{2}$, give $f(u) = r^2 + 2rs(f(e_\zeta) - \frac{1}{2})$, and $|e_\zeta\rangle\langle e_\zeta| + |e_{-\zeta}\rangle\langle e_{-\zeta}| = |\psi\rangle\langle\psi| + |\chi\rangle\langle\chi|$ gives $f(e_\zeta) + f(e_{-\zeta}) = 1$. If $d = f(e_\zeta) - \frac{1}{2} < 0$, the state with $r/s = -d$ has $f(u) = -d^2/(1+d^2) < 0$, so $f(e_\zeta) = \frac{1}{2}$ and $f(u) = r^2 = |\langle\psi|u\rangle|^2$.

Problems found in the brief. None that affect Theorem 9; some statements need care.

- EQST takes $S : \mathcal{E}(\mathcal{H}) \to \mathbb{R}$, but in infinite dimension the von Neumann entropy is $+\infty$ for some density operators, and $-\sum_i p_i \log p_i$ can be $+\infty$ for countably many weights. With a real-valued S, BR-ENT and [B]MUTEX cannot hold in infinite dimension, and Theorem 9 is true there only vacuously. The formalization uses $[0, \infty]$.
- The units are not fixed: eq. (2) writes log, while Corollary 7 and Fig. 2 use entropy 1 for two orthogonal states, which needs $\log_2$.
- BR-ENT calls $-\operatorname{tr}(\rho \log \rho)$ "the Shannon entropy".
- The proof of BR-ENT + [B]MUTEX ⇒ BR-BORN refers to "the proof of 1" and "the same trigonometric arguments in the proof of 1"; there is no item 1, and Proposition 3 and Corollary 7 are meant. It also says "p must take its extreme values on ψ and ϕ", where ψ and $\psi^\perp$ are meant.
- In the same proof, "Given that the range of p is $[0, 1]$, p must take its extreme values on ψ and $\psi^\perp$. Therefore, ψ and $\psi^\perp$ must be opposite points on the ball" does not say why the affine function $p(\psi|\cdot)$ has its gradient along the axis through ψ and $\psi^\perp$; that two orthogonal states are antipodal already holds in the Bloch ball of EQST. The missing step is that an affine function on the ball takes its maximum only in the direction of its gradient, so $p \leq 1$

with $p(\psi|\psi) = 1$ puts the gradient along ψ (`eq_of_chord_of_le_one`). The brief also does not say why $p(\psi|\cdot)$ is affine on the whole ball when EQST only makes it affine in mixtures of pure states: every segment of the ball lies on a chord, whose ends are pure states (`exists_linearMap_eq_of_chord`). With these two steps the argument of Fig. 3 is formalized as stated ("Differences", last item).